

Instructions for Project 2

CS 575

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Winter 2025

Abstract

The goal of Project 2 is to learn about complex contagions using two models of diffusion: a model that uses an *absolute threshold* and a model that uses a *fractional threshold*. You will experiment with several different kinds of fixed and canonical networks. You should interpret results in terms of network density, node centrality, local bridges, and network clustering. Finally, you will experiment with ways to choose early adopters to enhance the spread of a complex contagion.

1 Complex Contagion Model Elements

The spread of COVID-19 that you modeled in Project 1 occurred because a single infectious individual could infect any of its neighbors. In *complex contagion* models, called *diffusion* in [2], the spread of a disease, idea, behavior, or technology depends on the behavior of multiple neighbors.

In this project, you will explore how complex contagion spreads on various kinds of networks. Agents will have two states: (1) a **Baseline** behavior, belief, or technology, which is denoted by B and (2) an **Adopted** behavior, belief, or technology, which is denoted by A . Without loss of generality, this project refers to both the baseline and what is adopted as a *behavior*. An agent changes its behavior when enough of its neighbors adopt that same behavior.

There are many ways to model how agents behave. In this project, please use a simple finite state machine in your implementation.

1.1 State Initialization

The agent-based model has two states:

- State A: The agent has adopted behavior A
- State B: The agent has adopted behavior B .

Because the goal of the project is to help you understand how network structure affects the spread of a contagion, you will choose which agents begin the simulation using behavior A rather than behavior B . We call the agents who begin with behavior A *early adopters*. Early adopters begin in state State A. All other agents begin in state State B.

1.2 Transitions and Thresholds

You will explore two criteria under which agents will switch from the baseline behavior to the adopted behavior: *absolute threshold* and *fractional threshold*.

- *Absolute Threshold*. In the absolute threshold model, agents will switch if at least a of their neighbors have adopted the behavior. Experiment with $a \in \{2, 3, 4\}$.
- *Fractional Threshold*. In the fractional threshold model, agents will switch if at least a $f\%$ of their neighbors have adopted. Experiment with $f \in \{10\%, 20\%, 30\%$

Once an agent adopts behavior A , they never change.

1.3 Early Adopters

An early adopter is an agent that switches from the baseline behavior to the new behavior without requiring its neighbors to first adopt the new behavior. The location of early adopters in the network has a strong effect on how well the contagion spreads in the network.

The key idea for choosing early adopters is to understand the role of *local bridges* in the spread of complex contagion. Simply put, local bridges tend to block the spread of complex contagion because agents on the two sides of the bridge tend to have more neighbors on their side of the bridge than neighbors on the other side [3].

We learned from the Watts-Strogatz model that adding random edges between distant parts of the network induces the small-world property. These random edges tend to form local bridges precisely because they connect two disparate parts of the network. I hope you can see that the small world property makes simple contagion efficient because it allows the contagion to jump from one part of the network to another part.

Because the random edges in small-world networks tend to form local bridges, they tend to inhibit the spread of complex contagions. This is why complex contagions can spread efficiently within a part of the community but not broadly across network communities. (We'll formally define a network community in class, but we won't use it until Project 3.)

In addition to the specific early adopters described below, you will experiment with two ways of choosing early adopters: *clustered seeding* and *incubator neighborhoods*.

- *Clustered Seeding*. Clustered seeding means that you choose the early adopters that are neighbors of each other. The idea is that in networks with high clustering coefficient (remember triadic closure) choosing a bunch of adopters who are neighbors of each other will likely cause the adopters to “team up” on the other agents in their neighborhood.
- *Incubator Neighborhoods*. And incubator neighborhood is one where a handful of agents are chosen within the neighborhood in hopes that the complex contagion will spread efficiently within the neighborhood. If you add a few seeds in other neighborhoods, you could hope that when the contagion reaches a local bridge it might cross it because everyone on one side of the bridge has adopted and at least one person on the other has adopted.

Two specific quotes from [1] might provide more insight.

Practical suggestions for how to improve diffusion in these situations come in the form of two simple ideas: Clustered seeding can speed up the spread of a [complex contagion]. Clustered networks can create “incubator neighborhoods” that allow a challenger technology to outcompete an established competitor. In both cases, the main lesson is the same: increasing early adopters’ access to the network can backfire, leading to failed diffusion. Behaviors that face entrenched opposition can spread more effectively when early adopters have less exposure to the network.

The consequence for diffusion was that the underlying wide bridges in the spatial network were no longer able to provide enough reinforcement to overcome the countervailing influences from increasing numbers of weak ties. Thus, as neighborhood sizes increased, diffusion stopped. [p. 55]

2 Networks

There are nine networks with which you will experiment:

1. Figure 19.4 from Networks, Crowds, and Markets. See [Appendix A](#) for code to create the graph. Note that you will not report experiments for this network. It is included to help you build intuition.
2. a circulant graph with 20 vertices where each vertex is attached to two neighbors on either side (`nx.circulant_graph(20, [1,2])`)
3. a circulant graph with 20 vertices where each vertex is attached to four neighbors on either side (`nx.circulant_graph(20, [1,2,3,4])`)

4. **karate** network, which you can upload using the code in Appendix B.
5. a scale-free network with 100 vertices (`nx.barabasi_albert_graph(100,2)`)
6. a scale-free network with 410 vertices (`nx.barabasi_albert_graph(410,2)`)
7. a small-world network with 100 vertices `nx.watts_strogatz_graph(100,5,.3)`
8. a small-world network with 410 vertices `nx.watts_strogatz_graph(410,3,.3)`
9. **infect-dublin** network, which you used in Project 1.

The small-world networks are constructed using the Watts-Strogatz algorithm we discussed in class. Note that the algorithms for constructing the scale-free and small-world networks both use randomness, so you should run multiple trials with these networks and report average behavior.

3 Experiments to Conduct and Results to Report

The independent variables in this project are

- network type
- set of early adopters
- absolute threshold or fractional threshold

3.1 Specific Experiments

For both the absolute threshold and fractional threshold conditions, conduct the following experiments for given networks and early adopters

- Figure 19.4. *Do not report these results. They are suggested experiments to try to help you build intuition.*
 - Early adopters are vertices 1 and 2
 - Early adopters are vertices 7 and 8
 - Early adopters are vertices 6, 7, and 12
 - What nodes form the smallest set of early adopters that can guarantee that the new technology will cascade throughout the entire network?
- Circulant graph 20-2
 - Early adopters 1 and 2
 - Early adopters 1 and 10
- Circulant graph 20-4
 - Early adopters 1 and 2
 - Early adopters 1, 2, 3, and 4
 - What nodes form the smallest set of early adopters that will guarantee a complete cascade?
- Karate
 - Early adopters 1 and 2
 - Early adopters 33 and 34
 - What nodes form the smallest set of early adopters that will guarantee a complete cascade?
 - Which nodes are key to promoting a complete cascade? What is the clustering coefficient of those nodes?
- Small world 100

- The four nodes with lowest degree. *Feel free to share your code segment (on Discord) for finding the nodes with highest or lowest degree.*
- The four nodes with highest degree
- Scale free 100
 - The four nodes with lowest degree
 - The four nodes with highest degree
- Small world 410
 - The twenty nodes with highest degree
- Scale free 410
 - The twenty nodes with highest degree
 - The fifty nodes with highest degree
- Dublin
 - The fifty nodes with highest degree

3.2 Flexible Experiments

Use some creativity to choose early adopters for the `infect-dublin` network, the small-world networks, and the scale-free networks. Make specific hypotheses and then run experiments to test these hypotheses.

3.3 Dependent Variables

You have more flexibility in how you present results in this project than you did in Project 1. I suggest the following:

3.3.1 Time Dynamics for Small Graphs

For the smaller graphs (circulant 20-2, circulant 20-4, karate) consider presenting a sequence of screen shots that show how the cascade spreads over the network. If no cascade occurs, you can just say that. In my code, I set node color to blue for agents that used behavior *B* and node color to cyan for agents that used behavior *A*. My memory mnemonic was *B* for blue, and *A* for aquamarine (which is kind of like cyan).

3.3.2 Time Dynamics for Larger Graphs

For the larger graphs (small worlds, scale free, and dublin), plot the number of agents who have adopted behavior *A* as a function of time. For the small world and scale free graphs, the networks are formed with randomness, so you'll need to conduct multiple trials. Plot the mean and the interquartile range. For the dublin graph, only the mean is needed since there is no randomness.

3.3.3 Summarizing Results for Larger Graphs

For small world, scale free, and dublin graphs, figure out a way to summarize results. Please consider, for example, plotting the time until the cascade stops versus the clustering coefficient of the network. Similarly, you might consider the percentage of agents that adopted the new behavior as a function of network clustering coefficient. You might also consider similar scatter plots of time until the cascade stops and percentage of agents adopting as a function of other parameters like average degree, average path length, diameter, and graph density.

It's not a good idea to create all of those plots. You might include a table summarizing results, with one row per each of the large graphs and columns that might include average time-to-stop-changing, percentage adopting, clustering coefficient, graph density, average path length. If you present averages, please also include some indication of uncertainty for the small world and scale free graphs; standard deviation is easy, and so is value of upper and lower quartile.

4 What to Submit

Write a report of what you did and what you learned, and upload the report to Learning Suite. (A well-structured Jupyter notebook file can act as the report.) Please do not write more than four pages of text, but your figures and tables can make your report longer than four pages. You do not need to summarize the experiment conditions since I'm giving those to you. Please include the following:

- Names of everyone on your team. Please only submit one report per group on Learning Suite.
- Abstract: Summarize what you did and what you learned.
- Introduction: Short summary of why the problem is interesting, what you did, and how you organize your report.
- Hypotheses: Make some hypotheses that you think are interesting. Please try to make these before you see the results. I'd rather you form a hypothesis that ends up being uninteresting or wrong before you do the experiments than a clean hypothesis that is derived from the results.
- Discussion: Summarize which hypotheses were correct and which were incorrect. Explain why you think the results came out the way they did. If you are speculating about why, state that you are hypothesizing a possible explanation. If the reason why is justified by the data, tell me how the data supports your explanation. Since the existence or absence of clusters has a big impact on complex contagion, be sure your discussion includes something about clusters.
- Future Work: Tell me what you wish you had done or could do now that the project is over.
- You do not need a conclusion or summary.

A Method for generating graph from Figure 19.4

```
def _get_graph_nineteenfour_from_NCM_book():
    G = nx.Graph()
    G.add_nodes_from(range(1,16))
    G.add_edges_from([(1,2),(1,3),(2,3),(2,6),(4,5),(4,6),(4,7),(5,7),(5,8),(6,9),(7,8),(7,9),(7,10)]
    G.add_edges_from([(8,10),(8,14),(9,10),(9,11),(10,12),(11,12),(11,15),(12,13),(12,15),(12,16)])
    G.add_edges_from([(13,14),(13,16),(13,17),(14,17),(15,16),(16,17)])
    return G
```

B The karate network

Downloadable here:

<https://github.com/muvaf/CudaGraphCloseness/blob/master/karate.mtx>

References

- [1] Damon Centola. *How Behavior Spreads: The Science of Complex Contagions*. Princeton University Press, 2018.
- [2] David Easley, Jon Kleinberg, et al. *Networks, crowds, and markets*, volume 8. Cambridge university press Cambridge, 2010.
- [3] Douglas Guilbeault, Joshua Becker, and Damon Centola. Complex contagions: A decade in review. *Complex spreading phenomena in social systems: Influence and contagion in real-world social networks*, pages 3–25, 2018.